

Exploring Data Quality in the Verra VCS Registry

Exploratory Analysis on
Verra Energy Industry Public Data

Andita Rachmania · April 2026

1,700 Projects · Verra VCS Energy Industry

Why does registry data quality matter?

“ *Carbon-crediting programs shall provide comprehensive and transparent information on all credited mitigation activities ... accessible to non-specialised audiences, to enable scrutiny of mitigation activities.* ”

ICVCM Core Carbon Principles — Transparency Principle (2024)

The Integrity Council for the Voluntary Carbon Market (ICVCM) sets the global benchmark for high-quality carbon credits. Central to its framework is a Transparency Principle: project data must be public, structured, and auditable by independent parties.

Rating agencies like Sylvera and MSCI build their credit assessments on structured registry data. Where fields are missing or inconsistent, assessment becomes difficult — and so does informed decision-making.

About this analysis:

Applies data quality lenses to publicly available Verra VCS registry records, focusing on completeness, consistency, and structural patterns in the data.

Four datasets. One registry.

Verra VCS · Energy Industry · Exported March 26, 2026

1,700
All Projects

Complete universe — all statuses and lifecycle stages

1,387
Registered

Includes both Registered projects and Late to Verify (stalled) statuses

176
Pipeline

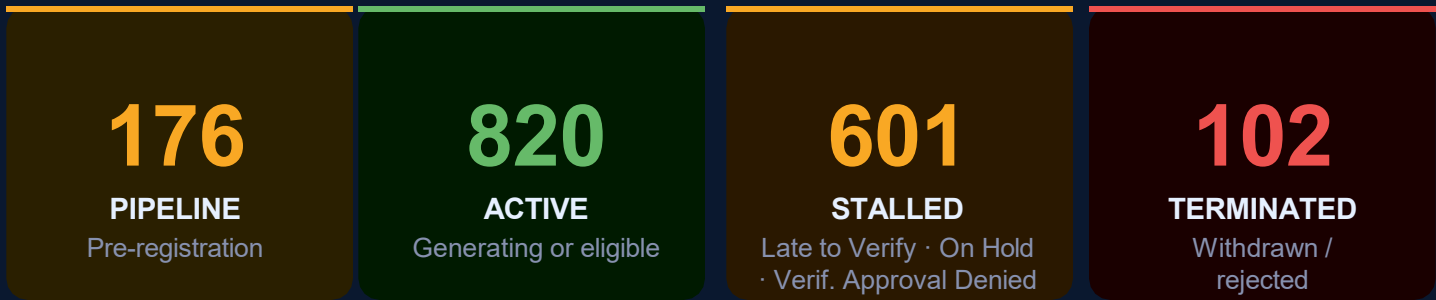
Pre-registration stage — projects under development or validation

102
Inactive

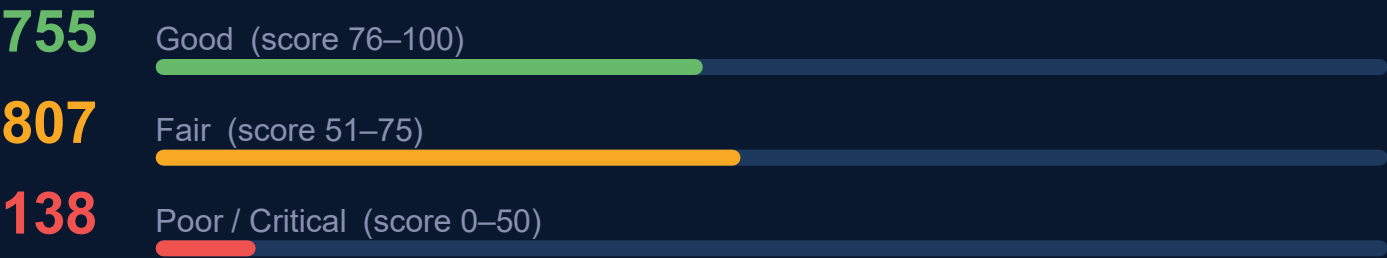
Terminal records — withdrawn, rejected, or declared inactive by administrator

Focus: data quality audit + anomaly detection across all 1,700 projects

1,700 projects. Four lifecycle stages.



Data quality score distribution



Quality score (0–100): field completeness weights

Methodology	20 pts
Registration Date	15 pts
Crediting Period Start Date	15 pts
Crediting Period End Date	15 pts
Region	10 pts
Estimated Emissions	10 pts
Project Name	5 pts
Proponent	5 pts
Valid Date Order (bonus)	+5 pts
Maximum score	100 pts

FINDING #1

The Zombie Registry

One status stands out among the 17 tracked in this dataset

The Verra VCS dataset tracks 17 distinct project statuses — from Under Development through Registered, Crediting Period Renewal, and Withdrawn. Among all statuses, one stands out in both volume and data pattern: “Late to Verify.”

597

projects carry a “Late to Verify” status

That’s 35.1% of the 1,700 projects in this dataset

35.1%

64.9% other statuses

What does “Late to Verify” mean in the data?

A project registered with Verra that has not recorded a completed verification cycle within the 5-year maximum monitoring window.

In the dataset: these records remain active in the registry but carry no verification outcomes — they are, in data terms, open records without a documented closure event.

FINDING #2

The 2020 CDM Migration Surge

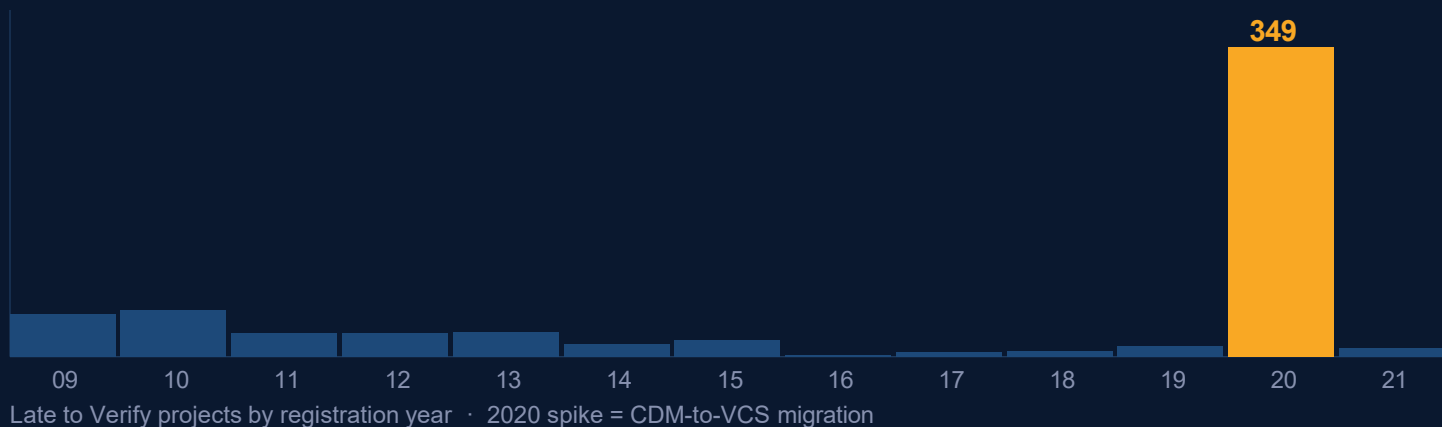
A registration year anomaly that shapes the entire dataset

58.5%

of all Late-to-Verify projects registered in a single year: 2020

What explains the 2020 spike?

The pattern is consistent with the sunseting of the UN's CDM (Clean Development Mechanism), which prompted migration of legacy projects to VCS standards. These 349 projects show data quality patterns typical of legacy record transfers.



FINDING #3

Data Quality Failures

Systematic gaps that undermine registry transparency

Most common missing fields

937 Crediting Period Dates
55% of projects

286 Registration Date
17% of projects

132 Region Classification
8% of projects

Risk level distribution & classification rules

HIGH 103	Score < 50 OR score < 75 with 2+ data flags	
MED 1,046	Score < 75 OR Late to Verify OR missing Crediting Period dates	
LOW 551	Score ≥ 75 No critical flags	

Where are the projects located?

Three countries hold 74% of all VCS energy projects. High-risk rates are broadly consistent.

Country	Projects	High-Risk count	High-Risk rate
India	652	39	6.0%
China	417	32	7.7%
Turkey	181	11	6.1%
Brazil	69	—	—
United States	41	—	—
South Africa	36	1	2.8%

High-risk rate = high-risk projects ÷ total projects per country

Observation:

High-risk rates (6–8%) are broadly in line with the dataset-wide average of 6.1%, suggesting that risk distribution is proportional to project volume, not geography.

What do the data patterns suggest?

01

Stalled Project Records

597 projects carry a 'Late to Verify' status, meaning their registry records have not been updated with verification outcomes. From a data standpoint, these are incomplete lifecycle records.

02

Legacy Data Gaps from CDM Migration

349 projects registered in 2020 show data quality patterns consistent with legacy records — incomplete fields, missing dates — that were carried over during a large-scale data migration.

03

Missing Dates Limit Analysis Depth

55% of projects are missing crediting period dates. For any analyst working with this dataset, these gaps constrain time-series analysis and lifecycle modelling without additional data sourcing.

Verra VCS Carbon Registry Analysis

Three patterns. One dataset.

Exploring 1,700 Verra VCS energy projects through a data quality lens surfaces patterns worth understanding before working with this data.

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of all projects are
"Late to Verify"

58.5%

of Late to Verify
projects registered
in 2020

55%

missing crediting
period dates

About this analysis

Conducted by Andita Rachmania — Data Analyst | Sustainability.

Dataset: Verra VCS Registry (Energy Industry), exported April 2026.

Explore further:

■ Interactive Dashboard

github.io/AnditaRachmania/verra-carbon-registry-analysis

■ Full EDA Notebook

github.com/AnditaRachmania/verra-carbon-registry-analysis